



2026

BIOLOGY

Grade 9 NCE | Paper 2

Time allowed: **45 minutes** Total marks: **50**

Candidate Name: _____

Index Number: _____

Class: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name, index number and class in the spaces provided above.
2. Answer ALL questions in Sections A, B and C.
3. For Section A, place a tick (✓) in the box next to your chosen answer. Only one answer per question.
4. Write your answers in the spaces provided in Sections B and C.
5. Use black or dark blue ink for all written answers. Use pencil for diagrams only.
6. The number of marks available is shown in brackets [] at the end of each question or part question.
7. Do not use correction fluid. Cross out any work you do not want to be marked.

Topics covered: Reproduction (Sexual & Asexual) • Human Circulatory System

For Examiner's Use Only

Section A /15	Section B /20	Section C /15	Total /50
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SECTION A — Multiple Choice Questions (15 marks)

Place a tick (✓) in ONE box for each question. Each question carries 1 mark.

- 1 Which type of reproduction produces offspring genetically identical to the parent? [1]
☐ A Sexual reproduction
☐ B Asexual reproduction
☐ C Fertilisation
☐ D Meiosis
- 2 Mitosis results in: [1]
☐ A Four haploid daughter cells
☐ B Two genetically identical diploid cells
☐ C Cells with half the chromosome number
☐ D Genetically varied offspring
- 3 Which of the following is an example of asexual reproduction in plants? [1]
☐ A Seed germination
☐ B Pollination
☐ C Runner formation in strawberry
☐ D Fertilisation
- 4 The male reproductive part of a flower is the: [1]
☐ A Pistil
☐ B Stigma
☐ C Carpel
☐ D Stamen
- 5 During meiosis, the chromosome number is: [1]
☐ A Doubled
☐ B Tripled
☐ C Halved
☐ D Unchanged
- 6 Which blood vessel carries oxygenated blood from the lungs to the heart? [1]
☐ A Pulmonary artery
☐ B Aorta

☐ C Pulmonary vein

☐ D Vena cava

7 The left ventricle pumps blood to the:

[1]

☐ A Lungs

☐ B Right atrium

☐ C Body (systemic circulation)

☐ D Left atrium

8 Which component of blood is responsible for clotting?

[1]

☐ A Red blood cells

☐ B White blood cells

☐ C Plasma

☐ D Platelets

9 Arteries differ from veins in that arteries:

[1]

☐ A Carry blood to the heart

☐ B Have thinner walls

☐ C Carry blood away from the heart under high pressure

☐ D Contain valves throughout their length

10 The chamber of the heart with the thickest muscular wall is the:

[1]

☐ A Right atrium

☐ B Left atrium

☐ C Right ventricle

☐ D Left ventricle

11 Which of the following is NOT a function of the blood?

[1]

☐ A Transport of nutrients

☐ B Gas exchange in cells

☐ C Fighting infection

☐ D Regulation of body temperature

12 Pollination is defined as the transfer of pollen from:

[1]

☐ A Anther to ovule

☐ B Stigma to ovary

☐ C Anther to stigma

☐ D Ovary to seed

13 A zygote is formed when:

[1]

- ☐ A Two body cells fuse
- ☐ B A sperm cell and egg cell fuse
- ☐ C A cell undergoes mitosis
- ☐ D Pollen lands on a stigma

14 The heartbeat is initiated by the:

[1]

- ☐ A AV node (atrioventricular node)
- ☐ B Bundle of His
- ☐ C SA node (sinoatrial node)
- ☐ D Purkinje fibres

15 Which blood cells carry haemoglobin to transport oxygen?

[1]

- ☐ A Platelets
- ☐ B White blood cells
- ☐ C Red blood cells
- ☐ D Lymphocytes

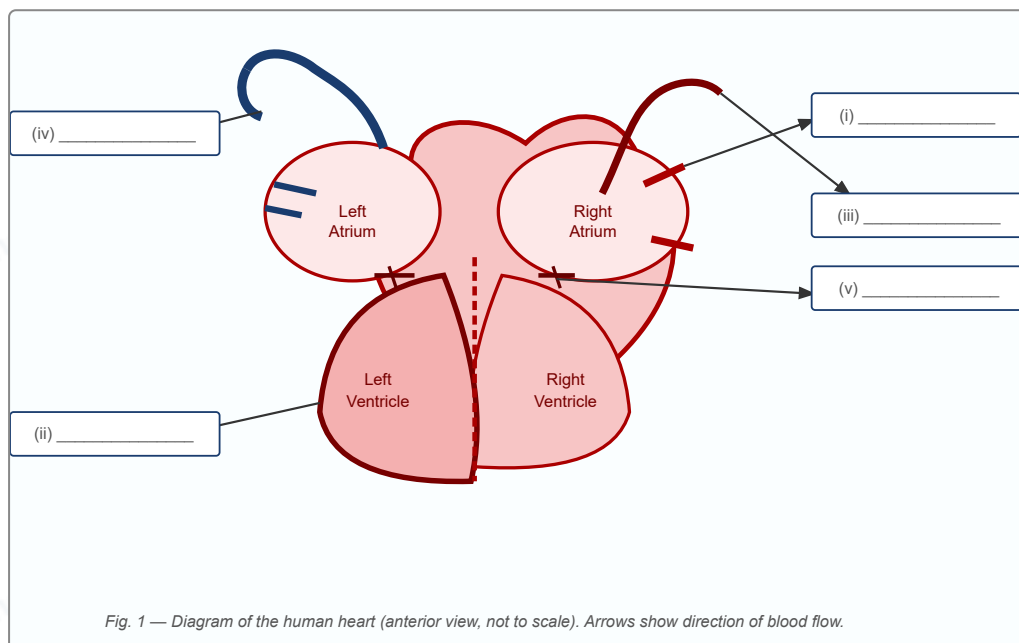
SECTION B — Structured Questions (20 marks)

Answer ALL questions. Write your answers in the spaces provided.

B1 Label the Heart Diagram

[5]

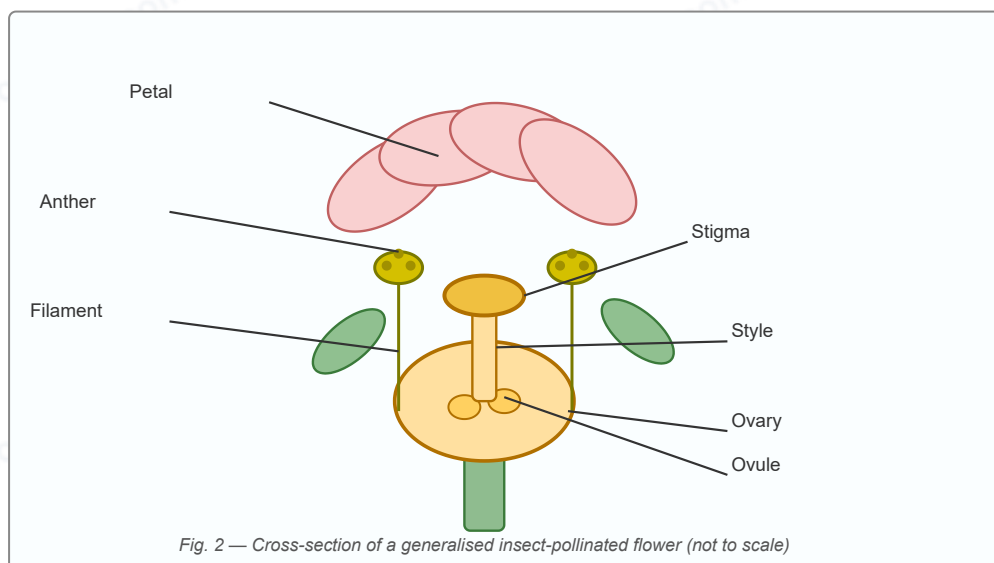
The diagram below shows the internal structure of the human heart. Write the correct label in each numbered box. Choose from: *Right atrium, Left ventricle, Pulmonary artery, Aorta, Tricuspid valve, Pulmonary vein, Left atrium, Right ventricle*.



B2 Flower Structure and Reproduction

[5]

The diagram below shows a cross-section through a typical insect-pollinated flower. Use the diagram to answer the questions that follow.



- (a) Name **TWO structures** visible in the diagram that are part of the **male reproductive system** of the flower.

[2]

1. _____
2. _____

(b) Describe what happens during **fertilisation** in a flowering plant. [2]

(c) State **ONE adaptation** of a wind-pollinated flower and explain how it helps pollination. [1]

B3 Mitosis and Meiosis**[5]**

- (a) State **TWO differences** between mitosis and meiosis.

[2]

Mitosis	Meiosis

- (b) Explain why meiosis is important for sexual reproduction.

[2]

- (c) In which organ of the human male does meiosis occur?

[1]**B4 Blood and Circulation****[5]**

- (a) State **TWO differences** between arteries and veins.

[2]

Arteries	Veins

- (b) Explain how red blood cells are adapted to carry oxygen efficiently.
Give **TWO adaptations**.

[2]

(c) What is meant by *double circulation* in mammals?

[1]

SECTION C — Extended Questions (15 marks)

Answer ALL questions.

C1 The Circulatory System

[8]

- (a) Describe the path of blood from the **right ventricle** back to the **left atrium** (pulmonary circulation). Name ALL the major structures the blood passes through in the correct order.

[4]

- (b) Explain what happens to the blood in the **capillaries** at the body tissues. Include the role of **plasma** and **lymph**.

[2]

- (c) A patient has a **blocked coronary artery**. Explain why this is dangerous and suggest **one treatment**.

[2]

C2 Reproduction in Plants and Animals

[7]

- (a) State **THREE advantages** of sexual reproduction over asexual reproduction.

[3]

1.

2.

3.

- (b) Describe the difference between **self-pollination** and **cross-pollination**. State **ONE advantage** of cross-pollination.

[2]

- (c) Vegetative propagation is a form of asexual reproduction. Describe **ONE method** of artificial vegetative propagation used by farmers and explain why it is useful.

[2]

MARK SCHEME — Biology Paper 2 (2026 Predicted)

This mark scheme is intended for **teacher / tutor use only**. Accept any answer that conveys the correct biological meaning, unless a question specifies a precise term. Where alternative wording is acceptable, it is indicated with *(accept:....)*.

SECTION A — Multiple Choice (1 mark each)

Q	Correct Answer	Marks
1	B — Asexual reproduction	1
2	B — Two genetically identical diploid cells	1
3	C — Runner formation in strawberry	1
4	D — Stamen	1
5	C — Halved	1
6	C — Pulmonary vein	1
7	C — Body (systemic circulation)	1
8	D — Platelets	1
9	C — Carry blood away from the heart under high pressure	1
10	D — Left ventricle	1
11	B — Gas exchange in cells (this happens at the tissues/cells, not in the blood itself)	1
12	C — Anther to stigma	1
13	B — A sperm cell and egg cell fuse	1
14	C — SA node (sinoatrial node)	1
15	C — Red blood cells	1
Section A Total		15

SECTION B — Structured Questions

Q	Answer / Marking Points	Marks
B1 (i)– (v)	(i) Right atrium (ii) Left ventricle (iii) Pulmonary artery (iv) Aorta (v) Tricuspid valve (accept: right atrioventricular valve) <i>Award 1 mark per correct label.</i>	5
B2 (a)	Any TWO from: Anther / Filament / Stamen / Pollen grains (accept if clearly referring to male parts visible in diagram). Award 1 mark each.	2

B2 (b)	A pollen grain lands on the stigma and a pollen tube grows down the style into the ovule (1). The male nucleus (from pollen) travels through the pollen tube and fuses with the egg cell (female nucleus / ovum) in the ovule to form a zygote (1).	2
B2 (c)	Any ONE valid adaptation with explanation, e.g.: Large / feathery stigma — to catch wind-borne pollen more effectively (1). OR: Pollen is small and light / produced in large quantities — easily carried by wind (1). OR: No petals / not brightly coloured — no need to attract insects (1).	1
B3 (a)	Any TWO valid differences (1 mark per contrasting pair): Mitosis: produces 2 cells / Meiosis: produces 4 cells Mitosis: produces diploid cells / Meiosis: produces haploid cells Mitosis: produces genetically identical cells / Meiosis: produces genetically varied cells Mitosis: occurs in body cells for growth and repair / Meiosis: occurs in gonads to produce gametes	2
B3 (b)	Meiosis halves the chromosome number to produce haploid gametes (1). When gametes fuse at fertilisation, the normal diploid chromosome number is restored in the offspring, maintaining the species' chromosome number (1).	2
B3 (c)	Testes (accept: testis)	1
B4 (a)	Any TWO valid contrasting pairs (1 mark per pair): Arteries: thick muscular walls / Veins: thinner walls Arteries: carry blood at high pressure / Veins: carry blood at low pressure Arteries: carry blood away from heart / Veins: carry blood towards heart Arteries: no valves (except at heart exit) / Veins: have valves to prevent backflow Arteries: narrow lumen / Veins: wider lumen	2
B4 (b)	Any TWO from: Biconcave disc shape — increases surface area for oxygen diffusion (1). No nucleus — more space for haemoglobin to carry more oxygen (1). Contains haemoglobin — binds readily with oxygen to form oxyhaemoglobin (1). Flexible / small — can pass through narrow capillaries (1).	2
B4 (c)	Blood passes through the heart twice in one complete circuit of the body: once through the lungs (pulmonary circulation) and once through the body tissues (systemic circulation) (accept any clear explanation showing two separate circuits both passing through the heart).	1
Section B Total		20

SECTION C — Extended Questions

Q	Answer / Marking Points	Marks
C1 (a)	Award 1 mark for each correct structure listed in the correct order (max 4 marks): Right ventricle → Pulmonary artery (1) → Lungs / capillaries of the lungs (1) → Pulmonary vein (1) → Left atrium (1). <i>All four structures must be named and in the correct sequence. Accept minor spelling errors if meaning is clear.</i>	4
C1 (b)	Oxygen and glucose diffuse from the blood in capillaries into the body cells; carbon dioxide and waste products diffuse from cells into the blood (1). Plasma (fluid) is forced out of the capillaries into the tissue fluid / lymph; this carries nutrients to cells. Excess tissue fluid is drained by the lymphatic system as lymph and returned to the blood (1).	2
C1 (c)	Danger: The coronary arteries supply the heart muscle with oxygen and nutrients. If blocked, the heart muscle is deprived of oxygen and cells begin to die, causing a heart attack (myocardial infarction), which can be fatal (1). Treatment: Any ONE of — coronary angioplasty (balloon / stent insertion) / coronary artery bypass graft (CABG) / medication to dissolve clot (thrombolytics) / statins to reduce cholesterol / lifestyle changes (accept any medically valid answer) (1).	2
C2 (a)	Any THREE from (1 mark each): 1. Produces genetic variation in offspring, helping the species to adapt to changing environments (1). 2. Variation means disease is less likely to wipe out the entire population (1). 3. New combinations of alleles can produce favourable traits (1). 4. Allows evolution through natural selection to occur over generations (1). <i>Accept any other biologically correct advantage.</i>	3
C2 (b)	Self-pollination: pollen is transferred from the anther to the stigma of the same flower (or another flower on the same plant). Cross-pollination: pollen is transferred from the anther of one plant to the stigma of a different plant of the same species. (1 mark for correct difference) Advantage of cross-pollination: produces greater genetic variation in offspring / reduces inbreeding / produces healthier offspring (1).	2
C2 (c)	Method: Any ONE of — cutting (stem / leaf / root cutting placed in compost to root) / grafting (taking a shoot from one plant and joining it to the rootstock of another) / budding / layering / use of tissue culture. (1 mark for naming/describing a valid method.) Usefulness: It produces large numbers of	2

	genetically identical offspring quickly / preserves desirable characteristics of the parent plant / does not require seeds so crop is uniform / faster than growing from seed (1).	
Section C Total		15
GRAND TOTAL		50

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